

SECTION 01001-A
GENERAL CONSTRUCTION REQUIREMENTS

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2.0 ADMINISTRATIVE REQUIREMENTS

- A. Cost Controls: Within 5 working days after award of the Contract, submit a resource-loaded or cost-loaded (as requested by the Owner), time-scaled CPM schedule that is to a level of detail that permits the accurate assessment of progress. The schedule shall take into account the coordination of scheduling requirements of other contractors working on the project. This schedule is subject to the approval of the Owner and shall be the project schedule upon approval. Adjustments must be made in a timely manner so as not to impact the project coordination. Include a program for updating and reporting progress against the approved schedule as set forth in Section 01001A.
- B. Preconstruction Meeting:
1. Within 5 days after the notice to commence construction or the start date specified in the Contract, the Owner will schedule and administer a preconstruction meeting with the Contractor and other contractors as Owner deems appropriate.
 2. The preconstruction meeting shall include representatives from the Owner, the Engineer, Contractor's project manager and field superintendents, and such other contractors, subcontractors, major suppliers, and others as Owner deems appropriate.
 3. The following items shall be included in the meeting agenda:
 - a. Designation of representatives for each party and their authority.
 - b. Scope of work review.
 - c. Construction schedule review, including delivery/storage of Owner-furnished equipment/material.
 - d. Facilities and equipment review, including specific items to be provided by the Owner, Engineer, or another contractor.
 - e. Paperwork requirements, including proof of workers' compensation insurance and certificates of liability insurance.
 - f. Design drawing transmittal and review process.
 - g. Shop drawing submittal process and schedule.
 - h. Proposal submittal process.
 - i. Extra work process.
 - j. Design clarifications and information requests process.
 - k. Progress billing process, including lien waivers.

- l. Connections to construction utilities.
- m. Special construction procedures requirements; e.g., cleanroom protocol.
- n. Parking and access requirements.
- o. Safety and security requirements, including accident reporting.
- p. Working hours of crews, including procedures for overtime.
- q. Work site work rules.
- r. QA/QC discussion.
- s. Use of client's facilities.
- t. Maintaining as-built documents.
- u. Hazardous materials control procedures.

C. Weekly Construction Meetings:

1. Provide an authorized representative to attend Owner's weekly construction progress meeting for the purpose of coordinating all phases of the work with other contractors and material suppliers, discussion of safety issues, and for preparation of detailed weekly work schedules including expediting requirements. To the extent commitments and schedules developed in these meetings do not constitute a change, they shall be binding upon the Contractor to the same degree as the approved scope and schedule.
2. Attendance by a Contractor's representative is mandatory when work is in progress.

D. Project Documents - Maintain at the work site one copy of the following Contract Documents available at all times for inspection by Owner:

1. The Contract, including addenda, amendments, field orders, change orders, and other modifications to the Contract.
2. Project specifications.
3. Project drawings provided by the Engineer, including additional drawings that may have been provided as a result of addenda, amendments, requests for information, clarifications, field orders, or change orders.
4. Reviewed shop drawings, product data, and samples.
5. Manufacturer's certificates, operation and maintenance manuals, warranties, inspection certificates and reports, field test records, permits, licenses, and other documents required by individual specification sections.

E. Work Limitations:

1. Perform work during normal working hours unless otherwise mutually agreed upon with the Owner.
2. Provide as little inconvenience as possible to Owner occupants of existing structures.
3. Schedule work in existing facilities and buildings, including Clean Zones, with the Owner sufficiently in advance to enable Owner to make arrangements as may be required.
4. Perform overtime work when defective work would result if deferred until the next regular-time work period and when emergency circumstances involving public welfare and safety or protection of property arise.
5. Cameras or use of cameras on the property without permission of Owner is prohibited.
6. Public Right-of-Way:
 - a. No roadway, street, alley, emergency fire lane, or public passageway shall be closed or obstructed except as required by the work and then only after approval of those having jurisdiction over same.
 - b. Use of streets and passageways shall be in accordance with directives from the Owner.
 - c. Private rights-of-way (e.g., Owner traffic patterns) shall not be interrupted without authorization from the Owner.

3.0 CONSTRUCTION ENVIRONMENT, HEALTH, AND SAFETY

A. Construction Safety Program:

1. Provide Contractor's Construction Safety and Health Program with the Bid including Contractor's policies, procedures, and practices that demonstrates:
 - a. Management commitment and leadership.
 - b. Assignment of safety responsibility.
 - c. Identification and control of hazards.
 - d. Employee education and training.
 - e. Incident analysis and record keeping.
 - f. Emergency planning and procedures.
 - g. Conformance with applicable Taiwan safety codes. Also comply with OSHA regulations (29 CFR Part 1926 or State program equivalent) as requested by the Owner including:
 - 1) Subpart A - General.
 - 2) Subpart B - General Interpretations.
 - 3) Subpart C - General Safety and Health Provisions.
 - 4) Subpart D - Occupational Health and Environmental Controls.

- 5) Subpart E - Personal Protective and Lifesaving Equipment.
- 6) Subpart F - Fire Protection and Prevention.
- 7) Subpart G - Signs, Signals and Barricades.
- 8) Subpart H - Materials Handling, Storage, Use, and Disposal.
- 9) Subpart I - Tools - Hand and Power.
- 10) Subpart J - Welding and Cutting.
- 11) Subpart K - Electrical.
- 12) Subpart L - Scaffolds.
- 13) Subpart M - Fall Protection.
- 14) Subpart N - Cranes, Derricks, Hoists, Elevators, and Conveyors.
- 15) Subpart O - Motor Vehicles, Mechanized Equipment, and Marine Operations.
- 16) Subpart P - Excavations.
- 17) Subpart Q - Concrete and Masonry Construction.
- 18) Subpart R - Steel Erection.
- 19) Subpart S - Tunnels and Shafts, Caissons, Cofferdams, and Compressed Air.
- 20) Subpart T - Demolition.
- 21) Subpart U - Blasting and Use of Explosives.
- 22) Subpart V - Power Transmission and Distribution.
- 23) Subpart W - Rollover Protective Structures, Overhead Protection.
- 24) Subpart X - Stairways and Ladders.
- 25) Subpart Y - Commercial Diving Operations.
- 26) Subpart Z - Toxic and Hazardous Substances.

- B. Project-Specific Safety Plan - Submit to Owner 10 days prior to beginning work, the Contractor's project-specific health and safety plan based on the preceding criteria in paragraphs 1.a through 1.g which coordinates with Owner's safety program, if any, addresses the safety criteria applicable to Contractor's work, and includes provisions for the following:
1. Contractor's designated, qualified, and authorized safety representative to supervise and enforce Contractor's compliance with the program.
 2. Contractor's Hazard Communication Program.
 3. Contractor's Project Field Instructions including:
 - a. Project information and description of major work phases.
 - b. Project site description.
 - c. Key personnel and emergency response contact information.
 - d. Emergency response arrangements.

- e. Job hazard identification and control measures.
- f. Personal protective equipment use specifications.
- g. Health hazard identification and control.
- h. Safe work procedures.
- i. Safe work practices.
- j. Employee safety education and training.
- k. Employee review and acknowledgement of the health and safety plan.

C. Hazardous Materials Control Plan:

- 1. Submit to Owner, prior to bringing hazardous materials on the work site but no later than 30 days after the notice to proceed, Contractor's Hazardous Materials Control Plan. This plan shall coordinate with Owner's environmental program, if any, address the criteria applicable to Contractor's work, and include provisions for the following:
 - a. List of hazardous materials and material safety data sheet for each item listed.
 - b. Personnel safety procedures to be instituted for handling hazardous materials.
 - c. Measures to be instituted for material storage, labeling, dispensing, accidental discharge, spill prevention, and containment.
 - d. Disposal methods for such materials.
 - e. Cleanup measures, materials, or structures, necessary to prevent environmental contamination by such material in the event of a spill or accidental discharge.
 - f. Name and contact information for Contractor's resources responsible for responding to an emergency involving hazardous materials.
- 2. Hazardous materials include any substances or mixtures which are:
 - a. Carcinogenic.
 - b. Toxic.
 - c. Corrosive.
 - d. An irritant.
 - e. Strong sensitizer.
 - f. Radioactive.
 - g. Explosive.
 - h. Pyrophoric.
 - i. Flammable.
 - j. Combustible.
 - k. Reactive.
 - l. Generate pressure through decomposition or heat.

D. Project Safety, Health, and Environment Requirements:

- 1. Contractor shall:

- a. Perform work in a safe manner, comply with safety, health, and environmental requirements of the Contract Documents as issued by the Owner and comply with applicable laws, codes, ordinances, rules, regulations, and lawful orders of public authorities.
- b. Attend, prior to beginning work, a project safety orientation conducted by Owner.
- c. Have and exercise, full legal responsibility for compliance to safety, health, and environmental rules and regulations by itself, its agents, employees, material suppliers, vendors, and subcontractors with respect to its portion of the work on the project; and directly receive and respond to, defend, and be responsible for any citation, fine, or penalty by reason of Contractor's failure or failure of Contractor's agents, employees, material suppliers, vendors, and subcontractors to so comply.
- d. Have a representative on the work site when a lower tier subcontractor is performing any work, even if no work is being performed by Contractor's direct work forces. This representative shall be responsible for assuring that applicable safety, health, and environmental rules and regulations are observed and in addition shall be responsible for responding to medical emergencies related to subcontractor's employees or sub-subcontractors.
- e. Uniformly enforce a policy that states the disciplinary action to be applied when employees violate safety and environmental rules. This policy will be consistent with Owner's safety and environmental enforcement policy, if any.
- f. Provide first aid and medical treatment for its employees.
- g. Provide safety and environmental training for its employees at its own expense, and document such training and provide copies to the Owner. Safety training shall include, but not be limited to:
 - 1) Orientation to the safety and environmental policies and rules stipulated by the Owner, if any, prior to each employee's initial work assignment on the project.
 - 2) Orientation of supervisors to the safety and environmental policies and rules and their responsibility to enforce them.
 - 3) Weekly toolbox talks on an appropriate safety and environment subject for all employees.

- 4) Hazardous materials training.
 - 5) Special safety training for those affected; such training shall include but not be limited to confined space entry, control of hazardous energy sources (electrical lockout/tagout), welding and cutting permit procedures, excavation and shoring, fall protection, scaffolding, etc.
 - 6) Specific instructions to its employees regarding the use of personal protective equipment, personal safe work practices, employee's obligations and rights under the Contractor's safety and environmental policies. The Contractor will provide at its own expense, required personal protective equipment for its employees and required safety equipment and supplies as needed.
- h. Conduct safety meetings with its employees and or supervision as necessary to ensure that safety and environment is adequately addressed in its work planning and execution. The Contractor will provide prior written notification of the time, place, and subject of the meetings and provide the Owner the opportunity to observe it. The Owner reserves the right to require changes necessary to comply with the project's safety rules and regulations.
 - i. Provide to the Owner copies of material safety data sheets (MSDS) for hazardous materials that Contractor brings on the work site, prior to the arrival of the materials on the work site. When requested by Owner, provide training to the Owner's employees and employees of other parties who may be routinely exposed to the materials used by Contractor. Such training will be at no additional cost to the Owner.
 - j. Provide a job safety analysis (JSA) identifying hazards before commencing each major phase or activity at the work site. The analysis will describe the plans and assign responsibility for hazard prevention or control of identified hazards.
 - k. Provide Contractor's safety representative to accompany Owner's safety representative during periodic inspections of Contractor's work and safety activities and take prompt action to correct identified deficiencies. Satisfactory compliance shall be made within a specified time. If Contractor refuses to correct deficiencies,

the Owner shall initiate appropriate actions in accordance with the Contract provisions and may take one or more of the following steps:

- 1) Cease the operation or a portion thereof (particularly in the case of an imminent danger).
 - 2) Correct the situation and back charge the Contractor.
 - 3) Stop or hold up payment for the work being performed.
 - 4) Invoke Contract penalties and/or terminate the Contract.
1. Attend a weekly site safety coordination meeting that will address results of periodic inspections, the safety or environmental related incidents, or injuries of the previous week, if any, and the necessary corrective measures including a plan for upcoming work and the related safety or environmental considerations.
 - m. Provide notification to the Owner as follows:
 - 1) Promptly report injuries or illnesses resulting in property damage and environmental incidents, and follow up in writing within 24 hours. Within 2 working days, provide a written report documenting the identified cause(s) of the accident or incident and action(s) taken, or planned to preclude recurrence. A copy of each first report of injury shall be provided.
 - 2) A weekly summary of occupational injuries, illnesses and environmental incidents including man-hours.
 - 3) Immediately report fatal or serious occupational injuries or illnesses. Notify the Owner in writing if an injury or illness results in lost work time and the Contractor desires not to provide continued employment, on a modified basis, for its employees who sustain an on-the-job partially disabling injury or illness. This notification must be provided a minimum of 24 hours prior to the discontinuance of employment. This reporting is in addition to any reporting responsibilities that the Contractor may have to local Taiwan regulation or any other agency. The Owner reserves the right to conduct an independent investigation at no additional cost to the Owner.

- 4) Immediately report Taiwan authority inspections and provide the Owner an opportunity to observe the inspection. Provide a copy of citations received and Contractor responses issued as a result of such inspections within two working days of receipt or issuance.
- n. Protect the safety and convenience of the general public including residents adjacent to the work.
 - 1) Perform work as necessary to protect the general public from hazards.
 - 2) Construct and maintain satisfactory and substantial barriers with warning lights, lanterns, and proper signs as necessary or required for safety.
- o. Ensure that these Contractor safety, health, and environment requirements are passed on to its subcontractors. The Contractor shall hold the subcontractor responsible for compliance with these requirements.
- p. Appoint an employee who is qualified and authorized to supervise and enforce compliance with the safety program.
2. The duties of the Owner or Engineer do not include a review or approval of the adequacy of the Contractor's safety supervisor, the safety program, or safety measures taken in, on, or near the construction site.
3. The "clean and safety requirements" are part of the Contract. The Contractor shall follow the requirement.

4.0 WORK SITE SECURITY

- A. Security is the responsibility of the Contractor until acceptance of work by the Owner. The Owner and Engineer do not assume responsibility at any time for the protection of the building and premises or for the loss of materials and/or equipment from the time that Contractor's operations have commenced until final acceptance of the work. Provide necessary security program and facilities to protect work from unauthorized entry, vandalism, and theft.
- B. No employee of Contractor, nor of its subcontractors, shall go to or be present in an area where the employee is not actively engaged in work, except when traveling between work location. Such travel shall be by a direct route. Use of any Owner facilities such as cafeteria,

office space, restrooms is prohibited. Contractor's employees will not be permitted to use Owner's parking facilities and shall park in areas designated by the Owner.

- C. Do not drive privately owned or leased vehicles inside the work site fenced area. Work site access passes for authorized vehicles will be limited and will be restricted to company vehicles. Passes will be issued by Owner and will be subject to applicable rules and regulations.
- D. Workers or visitor identification badges must be prominently displayed by all personnel when entering, working on, or leaving Owner's premises. Issue a badge to every employee or visitor when entering the site and collect the badge upon departure. Badges for workers will be picture identification badges issued by Owner designating the employer's name and the employee's identification number. Provide a current index showing each authorized employee when requested by Owner. Advise Owner when an employee is terminated from the project and no longer have access to the work site.
- E. Ensure that employee hard hats clearly identify the employer's company.

5.0 QUALITY CONTROL

- A. Develop and maintain a quality control (QC) manual as described in Section 01001B to support compliance with the Contract Documents. Maintain quality control over suppliers, manufacturers, products, services, work site conditions, and workmanship to produce work of specified quality.

6.0 COORDINATION

- A. Coordination:
 - 1. Cooperate with the Owner, other contractors, and utility companies on site in scheduling and coordination of activities in a manner that will provide the least interference of others' work and cooperate in the interfacing and connection of the separate elements of the overall project. The Owner and other contractors have the same obligation to cooperate and coordinate their work with Contractor. The Owner will issue such instruction as necessary. Abide and support the Owner's decision.

2. Coordinate the planned interruption of any services including, but not limited to, water, gas, oxygen, electric lights or power, and drainage systems, with the Owner. Advise Owner of the time, duration, extent, and type of any planned interruptions. Provide detailed plan in accordance with Owner's safety and operation and maintenance procedures. Make such interruption only after Owner has approved the plan.
3. Coordinate work of the various sections of the Specifications and Drawings to ensure efficient and orderly sequence of installation of construction elements with provisions for accommodating items installed later.
4. Locations of existing utility mains, trees, etc. are shown diagrammatically on the Drawings. Accuracy and completeness of this information is not guaranteed. Verify the size and location of utility lines, conduit, surface, or subsurface structures, etc., that may affect the work.
5. Coordinate work with other contractors and subcontractors and share information with the other contractors to enable you and them to coordinate openings, relationship to structure, ducts, pipe, conduit, and other items as required.
6. Coordination meetings are mandatory in laying out pipe and duct runs and equipment locations. Review these layouts with other contractors, the Engineer, and the Owner prior to installation.
7. Submit proposed method of support for review by the Engineer prior to fabrication of hanging or other support systems not specifically detailed on the Drawings.
8. Verify that characteristics of elements of interrelated operating equipment are compatible; coordinate work with other contractors having interdependent responsibilities for installing, connecting to, and placing in service such equipment.
9. Do not endanger the work of other contractors by cutting, excavating, or otherwise altering their work. Prior to penetrating load-bearing walls or structural members not detailed in the Contract Documents, submit written notice thereof and obtain permission from the Owner and Engineer before such work can proceed.
10. Coordinate space requirements and installation of mechanical and electrical work that is indicated on the Drawings. Follow routing shown for pipes, ducts, and conduit as closely as practicable; make runs

parallel with lines of building. Utilize spaces efficiently to maximize accessibility for other installations, maintenance, and repairs.

11. Provide clear access to equipment spaces. Do not obstruct placement or removal of equipment with conduit, ducts, utilities, or other facilities.
 12. In finished areas, except as otherwise shown, conceal pipes, ducts, and wiring in the construction. Coordinate locations of fixtures and outlets with finish elements.
 13. During the progress of construction, it is expected that minor field relocations of the work will be necessary. Such relocations shall be made only by direction of the Owner. If the Contractor shall fail to so notify the Owner and proceed without such direction, Contractor shall do so at its own risk.
- B. Cutting and Patching:
1. Cut, fit, or patch work as required to make its several parts fit together properly, to uncover work, to provide for installation of ill-timed work, to remove samples of work as specified for inspection and testing, and to install work in existing construction.
 2. Except as specifically permitted in the Contract Documents, do not cut or alter the work of another contractor except with the written consent of the Owner.
 3. Coordinate requirements of openings for ducts, piping, conduit, and other penetrations required by construction with the Owner. Unless otherwise specified, such openings shall be provided by the contractor that performs the initial construction. In the event of the Contractor's failure to supply the Owner with information regarding the size and location of openings it requires in advance, cutting and patching that may be required shall be done at the expense of the negligent party.
 4. Unless otherwise specified by the Owner, each contractor shall do cutting, fitting, or patching of its work that may be necessary to make said work coordinate with the work of other contractors, as shown or reasonably implied from the Contract Documents. In cases where the completion of the work of the Contractor is dependent upon work of various contractors, the timing of the work of each contractor will be determined by the Owner.
 5. Seal penetrations through nonfire-rated construction with permanently flexible material per Engineer's

specification or approval,. Oversized openings in nonrated construction, where the annular space around penetrants exceeds 1 foot, shall be closed by the contractor responsible for the initial construction, at the expense of the responsible party.

6. Close penetrations through or between fire rated walls, floors, and roofs in accordance with Engineer's specification or approval. Provide sleeving, coring, or cutting of fire-rated construction as described in the approved, UL-listed firestop assembly. Sleeving, cutting, or coring not provided as so described shall be corrected at no cost to the Owner. Installation of firestopping material shall be performed using an approved method or system and by personnel approved by the manufacturer.
7. Permission to patch or repair damaged areas or items of work shall not constitute a waiver of the Owner's right to require complete removal and replacement of work if, in the Owner's opinion, the patching or repairing does not satisfactorily restore the quality or appearance of the work.

7.0 TEMPORARY FACILITIES AND CONTROLS

A. Temporary Construction Facilities:

1. Before moving onto the work site, submit for approval a layout of the site showing location of security fencing, office areas, storage areas, workshops, and other temporary facilities consistent with Owner guidelines.
2. Confine operations at the work site to areas permitted by law, ordinances, permits, and the Contract Documents. Provide and locate or erect temporary buildings (change houses, trailers, plan rooms, offices, etc., but not major storage yards, layout and fabrication yards, equipment repair facilities, or Contractor personnel parking) within the indicated limits of construction with the approval of the Owner. Such temporary buildings, together with the necessary utilities, shall remain the property of the Contractor.
3. Do not unreasonably encumber the work site with materials or equipment. Operations of the Contractor (including storage of materials and equipment upon the work site) shall be confined to the area shown on the Drawings or where otherwise approved in writing by the Owner. No unauthorized or unwarranted entry, passage through, or storage or disposal of materials

shall be made upon the premises. Move stored products that interfere with the operations of other subcontractors.

4. Confine traffic to the rights-of-way of the work site and repair damage resulting from Contractor traffic to the satisfaction of the Owner. Road usage must be shared and coordinated among all users. Use assigned gates only. Always maintain Owner's access to its property. Maintain your temporary laydown areas. Keep work areas, walkways, roadways, and laydown areas that are under your control free from snow and ice.
 5. Locate major storage yards, layout and fabrication yards, equipment repair facilities, etc., in space assigned by the Owner. Provide temporary improvements including, but not limited to, site drainage, fencing, lighting, electric power, etc., as required. Obtain and pay for the use of additional offsite storage or work areas needed for the Contractor's construction activities.
 6. Provide break area for Contractor's personnel including adequate waste containers for food waste and garbage. Instruct personnel and subcontractor personnel that Owner's vending machines are not to be used, and take other necessary action to ensure that instructions are followed. Provide vending machines near break area as desired.
- B. Temporary Utilities:
1. Electricity and Lighting:
 - a. Provide electricity source necessary for construction activities. Provide source connection of temporary electrical power from power panels including necessary extension cords, transformers, and protective devices.
 - b. Owner will provide and but not pay for temporary electrical power from those sources, except those requirements beyond the capability of Owner's permanent system and the sources described above. Electrical requirements such as welders and requirements too far removed from the source will be Contractor's responsibility.
 - c. Temporary installations shall meet applicable local codes, regulations and other requirements. Do not plug into permanent receptacles without specific permission or instruction from Owner. Any unplanned service interruption on Owner's facility is unacceptable.

- d. If the permanent electrical power system is not available for use at the time of Contractor's mobilization, provide electrical power required by its operations until such time that Owner authorizes connection and use of the permanent electrical power system. For this purpose, provide an alternative mobilization cost for establishing the means to provide temporary electrical power and an alternative daily cost for providing and maintaining temporary electrical power until such power is available from the permanent system.
 - e. Provide lighting necessary for construction activities and to satisfy safety and security requirements.
2. Heat and Ventilation:
- a. Provide as required to maintain specified conditions for construction operations, worker comfort, and to protect materials and finishes from damage due to temperature or humidity. Provide power to temporary units in a safe and efficient manner until an acceptable power source is available from others.
 - b. Provide ventilation of enclosed areas to cure materials, to disperse humidity, and to prevent accumulations of dust, fumes, vapors, or gases.
 - c. Associated costs shall be paid by the Contractor.
3. Water: Water service does exist on site.
- a. Construction Water: Provide service required for construction operations. Provide temporary connections as required including hoses, nozzles, buckets, etc. Observe good housekeeping practices in the area of the connection point at all times.
 - b. Potable Water: Furnish containers, ice, distribution system, etc., for potable water as required for Contractor's personnel and its field office. No Contractor personnel (or any of its subcontractors' employees) are permitted to use permanent wash rooms, drinking fountains, or locker rooms of the Owner.
4. Sanitary Facilities: Temporary toilet facilities will be provided and maintained by others at a central location as determined by the Owner. Facilities include portable toilets and wash facilities. Require the usage of these facilities in lieu of Owner facilities by Contractor personnel.
5. Waste Management: Provide for disposal of garbage and debris in accordance with regulatory and Owner

requirements. Provide and enforce the use of proper receptacles for disposal of food wrappers and containers, empty daily. Segregate food related trash from construction-related trash and debris. Provide a written waste management plan within 30 days of award of the Contract covering handling and disposal of all wastes. Implement the plan as approved by Owner. Include the following details in the plan.

- a. Identify which of the following wastes will be recycled, either through recycling contractors, scrap yards, materials distributors, or manufacturers: metals, drywall, concrete, asphalt, cardboard, carpet, wood, clean fill, or others.
 - b. Identify wastes that will be taken to landfill.
 - c. Describe whether material to be recycled will be separated from other wastes on site by Contractor or off site by waste hauler or landfill operator. If separated off site, provide name and phone number of off site operator. If separated on site, describe the number and location on site of drop boxes for each type of material separated.
 - d. Identify materials or equipment that will be sold directly for salvage.
6. Telephone Services: Arrange and pay for individual telephone and fax service, if desired, at the field office by Contractor.
 7. Gases: Provide compressed air, oxygen, welding gases, and other expendables required in the performance of work by Contractor.
- C. Protection of Property and Risk of Loss:
1. Until final acceptance of work, the Contractor shall be responsible for the care, custody, control, safekeeping, and preservation of work and materials, equipment, supplies, and other things relating to work furnished by the Owner. Promptly repair or replace such if damaged or lost.
 2. Provide temporary protection for installed products. Continuously maintain adequate protection of work and work in progress from loss and damage in a manner satisfactory to the Owner. Work and work in progress includes goods, materials, and equipment furnished by others, whether in storage on or off the work site or otherwise handled in a way that is under the care, custody, or control of the Contractor or its subcontractors.

3. Place and operate construction equipment, tools, and facilities in such a manner that new and existing work will not be damaged by vibration, smoke, dust, water, falling debris, concrete, or any other cause. Protect sidewalks, steps, and ramps from heavy equipment traffic. Repair damage to roads, walks, and structures caused by your equipment or personnel.
4. Adequately protect adjacent private and public property as provided by terms of permits, applicable laws, rules and regulations, and the Contract Documents. Provide temporary protection of occupied and operating areas of Owner's existing plant or facilities. Protect other property at the work site or adjacent thereto, including, but not limited to, trees, shrubs, lawns, walks, pavements, roadways, structures, and utilities from damages caused by work.
5. Provide barriers around trees and plants designated to remain. Protect against vehicular traffic, stored materials, dumping, chemically injurious materials, and puddling or continuous running water.
6. Provide protection of work from freezing, rain, wind, and other elements of the weather that would be harmful to it. Furnish fuel, protective shelters, or temporary buildings and temporary heating as required for the prosecution and protection of work and keep work areas free of water. Coal and kerosene-type salamander stoves, pots, and open fires will not be permitted.
7. Provide protection coverings at walls, projections, jambs, sills, and soffits of openings. Protect finished floors and stairs from traffic, movement of heavy objects, and storage.
8. Prohibit traffic and storage on waterproofed and roofed surfaces, lawn, and landscaped areas. When unavoidable, protect surfaces from puncture, burn, or other damage when working above or on such surfaces. Pick up screws, nails, metal scraps, and other objects that may damage the surface.
9. Provide shoring, bracing, and supports as necessary to maintain the structural integrity of the project to the extent threatened or potentially threatened by work. Excavating required shall be performed by methods that will prevent damage to other work. Provide and maintain adequate temporary supports, shores, and bracing as necessary to maintain the structural integrity of the project and to keep work safe from failure or damage due to loads that may be imposed on structures during construction. Excavating required shall be performed by methods

that prevent damage to other work or existing facilities.

10. Do not load structures with weight that will endanger it.
 11. Provide barriers as required to prevent public entry to construction areas and to protect existing facilities and adjacent properties from damage from construction operations. Control traffic in immediate area to minimize damage.
 12. Provide temporary, weathertight closures of openings in exterior surfaces to provide acceptable working conditions and protection for materials to allow for temporary heating and to prevent entry of unauthorized persons. The temporary closures shall be maintained until permanent doors and windows are in place and operational.
- D. Water and Erosion Control:
1. Provide methods to control surface water to prevent damage to the project, the site, or adjoining properties. Control fill, grading, and ditching to direct surface drainage away from excavations, other construction areas, and to direct drainage to proper drainage facilities.
 2. Provide, operate, and maintain equipment of adequate capacity to control surface water and groundwater as necessary.
 3. Dispose of drainage water in a manner to prevent flooding, erosion, or other damage the site or to adjoining areas.
 4. Plan and execute construction earthwork methods to control surface drainage from cuts and fills and prevent erosion and sedimentation.
 - a. Rough grade site to prevent standing water and to direct surface drainage away from excavations, adjoining properties, and public right of way.
 - b. Hold the areas of bare soil exposed at one time to a minimum.
 5. Construct fill areas by selective placement to eliminate surfaces which will erode.
- E. Pollution and Dust Control:
1. Provide methods, means, and facilities required to prevent contamination of soil, water, or the atmosphere by the discharge of noxious substances from Contractor's construction operations.
 2. Provide equipment and personnel and perform emergency measures required to contain spillage and to remove contaminated soils or liquids. Excavate and dispose

- of contaminated earth off site and replace with suitable compacted fill and topsoil.
3. Take special measures to prevent harmful substances from entering public waters. Prevent disposal of wastes, effluents, chemicals, or other such substances adjacent to ponds or in sanitary or storm sewers.
 4. Equipment or material used for the construction of the project that produces toxic fumes may be used at times specified by the Owner.
 5. Screen noisy equipment with temporary enclosures to shield adjacent areas as much as possible.
 6. Provide scrubber devices on the exhaust of gas driven equipment used inside a building or other enclosed space.
 7. Provide positive methods and apply dust control materials to minimize raising dust from construction operations and provide positive means to prevent airborne dust from dispersing into the atmosphere.
 8. Burning or burying of waste materials on the work site is prohibited.

8.0 EXECUTION REQUIREMENTS

A. Site Investigation and Representation:

1. Contractor acknowledges satisfaction as to the nature and location of work; the general and local conditions, particularly those bearing upon availability of transportation, access to the work site, disposal, handling, and storage of materials, availability of labor, water, electric power, and roads, telephone communications waste and sewage disposal and uncertainties of weather or similar physical conditions at the work site; the conformation and conditions of the ground; the character of equipment and facilities needed preliminary to and during the prosecution of work; and other matters that can affect work or the cost thereof under this Contract.
2. Contractor further acknowledges satisfaction as to character, quality, and quantity of surface and subsurface materials to be encountered from its inspection of the work site and from reviewing available records of exploratory work furnished by the Owner or included in the Contract Documents. Failure by Contractor to become acquainted with the physical conditions of the work site and the available information will not relieve the Contractor

from responsibility for properly estimating the difficulty or cost of successfully performing work.

B. Contractor's Responsibility for Utility Location:

1. Known utilities and structures adjacent to or within the limits of work are shown on the Drawings. The locations shown are taken from existing records and the best information available from existing utility plans; however, it is expected that there may be some discrepancies and omissions in the locations and quantities of utilities and structures shown.
2. Where Contractor's operations could cause damage or inconvenience to telegraph, telephone, television, power, oil, gas, water, sewer, or irrigation systems, suspend operations until Contractor has made arrangements necessary for the protection of these utilities and services. Employ or secure services as necessary to locate or verify the location of utilities.
3. Notify utility offices that could be affected by work at least 48 hours in advance. Do not expose utilities without first obtaining permission from the appropriate agency. Once permission has been granted, locate, expose, and provide temporary support for existing underground utilities.
4. Neither the Owner nor its officers or agents shall be responsible to Contractor for damages as a result of Contractor's failure to locate and protect utilities encountered in work.
5. Tag stubouts for underground utilities such as plumbing, mechanical, automatic fire sprinkler, and water with weatherproof identification tags.

C. Layout Data:

1. Baselines:
 - a. The Owner will provide two reference baselines and a single permanent benchmark to be used by the Contractors for layout work. Verify and agree to the accuracy of this data prior to commencing any layout work.
 - b. Notify Owner, in writing, if an apparent error in survey locations furnished by Owner is discovered. Request resolution. In no case shall the Contractor claim delay due to an apparent error to Owner-furnished locations except for instances in which the Owner failed to provide prompt direction upon receipt of Contractor's notice of an apparent error.

- c. The responsibility for the accuracy of the layout of work obtained from these baselines and benchmarks is solely that of the Contractor.
 - d. Furnish, at Contractor's expense, stakes, spikes, steel pins, templates, platforms, equipment, tools, and materials and labor required to lay out the work from the benchmark and baselines established by Owner.
 - e. Maintain and preserve stakes and other markers established by Contractor until no longer required.
 - f. Replace, at Contractor's expense, the control points established at the site by Owner, if destroyed by or through Contractor's negligence prior to their authorized removal. Owner may require work to be suspended when horizontal or vertical control points established at the work site by Contractor are not adequate to permit checking the work. Such suspension will be withdrawn upon proper replacement of the control points.
- 2. Record Contractor's survey data in accordance with standard and approved methods. Make available field notes, sketches, recordings, and computations made by the Contractor in establishing horizontal and vertical control points during the progress of work for ready examination by the Owner.
 - 3. Dimension Verification:
 - a. Before ordering material or doing work, take or verify measurements as may be required for the proper fitting of Contractor's work or other adjoining work. The Contractor is responsible for the correctness of its figures and shall satisfactorily correct work that does not fit, and furnish new work, if necessary, without charge to the Owner. Extra charges will not be allowed on account of differences between actual dimensions and the measurements indicated on the Drawings.
 - b. In case of a discrepancy or conflict with given data, notify Owner in writing prior to proceeding with the affected work.
- D. Preservation, Restoration, and Cleanup:
- 1. Keep the work site orderly and neat on a daily basis. Maintain a level of cleanliness acceptable to Owner. Rubbish shall not be allowed to accumulate anywhere on the work site.

2. Provide for regular maintenance of temporary toilets and replacement of waste containers when full.
3. Keep the paved streets bordering the work site clean from material tracked off site. Wash and/or sweep the streets daily if material begins to accumulate.
4. Smoking and eating will be allowed on the work site only in designated areas. Owner will identify the areas after award of the Contract.
5. Remove temporary materials, equipment, services, and construction prior to substantial completion inspection. Clear areas used by the Contractor for temporary structures, rubbish, and waste materials and properly grade to drain and blend in with the abutting property. Finish areas used for the deposit of waste materials to properly drain and blend with the surrounding terrain. Clear and restore staging areas to the original condition when the Contractor moved onto the work site. Correct damage, changes, or alterations, made by the Contractor to storage or laydown areas during construction prior to demobilization. Areas shall be returned to the condition that existed prior to arrival at the work site.
6. Clean and repair damage caused by installation or use of temporary facilities. Remove temporary underground installations. Grade site to drain.
7. Replace filters and clean system internals if permanent air-handling support equipment is used prior to acceptance by the Owner.
8. Thoroughly clean parts of the installation at the completion of work. Clean up and remove from the premises refuse material, crates, and rubbish arising from work.
9. Remove temporary drainage and erosion protection systems when no longer required.

9.0 CLOSEOUT PROCEDURES

A. General:

1. This Section contains requirements and procedures for submittal of pertinent data relating to closing out work upon completion of work. Detailed instructions elsewhere in these Specifications may require that certain items listed herein be submitted prior to the Substantial Completion date. Receipt and review of items specified in this Section are prerequisites for final payment.
2. At the completion of work, the site occupied by the Contractor shall be brought back to its original

condition. With the written consent of the Owner, improvements made by the Contractor may be abandoned and not removed and will then become the property of the Owner.

B. Semifinal and Final Inspections:

1. When the Contractor is of the opinion that the project is substantially complete, request inspections as required by the Contract terms. Such notice shall be given at least 3 working days before the requested inspection date.
2. If because of acts or omissions of the Contractor, the Engineer is required to conduct more than one final inspection of the project, the Engineer may charge the Owner for the additional services required, and such costs will be deducted from the money still due the Contractor.
3. Submit the following project documents to the Owner for review by the Engineer prior to completing the punch list of incomplete or uncorrected items:
 - a. Required operation and maintenance manuals.
 - b. Warranties and bonds.
 - c. As-built drawings.
4. When all items on the punch list are completed and corrected, send to the Owner a statement that the project is complete and request a final inspection. An acceptable final inspection by Owner is required before acceptance and final payment.
5. Schedule and conduct the final inspection at a time arrived at with the Owner. The final inspection shall include a complete checkout and test of mechanical and electrical systems, architectural and structural devices, etc. For this purpose, each trade concerned shall provide a skilled operating engineer or technician for a period of at least 1-day. Test all systems and devices and demonstrate the complete operation and required maintenance of each.
6. Make changes to the Contract Documents identified by the Owner and Engineer and resubmit the final documents.

C. Contract Closeout:

1. At the conclusion of work, turn over to Owner and obtain signed receipts for keys to lockable doors, panels, gates, controls, etc., properly identified and tagged.
2. Submit final application for payment in accordance with the Contract.

3. Complete and deliver to the Owner the following items (as applicable) prior to final payment.
 - a. Verification of completion of the punch list.
 - b. Fully executed certificate of Substantial Completion.
 - c. Local, state, or federal inspections and/or certifications.
 - d. 1-year workmanship and materials guarantee.
 - e. Roofing guarantee.
 - f. Wood door guarantee.
 - g. Major mechanical/electrical equipment guarantees.
 - h. Operation and maintenance manuals (six sets).
 - i. Project as-built documents.
 - j. Contractor lien release.
 - k. Consent of surety.
 - l. Response to field orders.
 - m. Change orders fully executed.
 - n. Approval of final pay application.
 - o. Spare parts and maintenance materials.

- END -